

torch.nn.functional.cross_entropy#

https://pytorch.org/docs/stable/generated/torch.nn.functional.cross_entropy.

Description:

Compute the cross entropy loss between input logits and target. See `CrossEntropyLoss` for details. `input` (Tensor) – Predicted unnormalized logits; see Shape section below for supported shapes. `target` (Tensor) – Ground truth class indices or class probabilities; see Shape section below for supported shapes. `weight` (Tensor, optional) – a manual rescaling weight given to each class. If given, has to be a Tensor of size C

Function Signature

```
torch.nn.functional.cross_entropy(input, target,  
weight=None, size_average=None, ignore_index=-100,  
reduce=None, reduction='mean', label_smoothing=0.0)[source]#
```

Parameters

Return type

Shape:

Returns:

Tensor

Code Examples

```
>>> # Example of target with class indices
>>> input = torch.randn(3, 5, requires_grad=True)
>>> target = torch.randint(5, (3,), dtype=torch.int64)
>>> loss = F.cross_entropy(input, target)
>>> loss.backward()
>>>
>>> # Example of target with class probabilities
>>> input = torch.randn(3, 5, requires_grad=True)
>>> target = torch.randn(3, 5).softmax(dim=1)
>>> loss = F.cross_entropy(input, target)
>>> loss.backward()
```

Detailed Documentation

Documentation

Compute the cross entropy loss between input logits and target.

See `CrossEntropyLoss` for details.

input (Tensor) – Predicted unnormalized logits; see Shape section below for supported shapes.

target (Tensor) – Ground truth class indices or class probabilities; see Shape section below for supported shapes.

weight (Tensor, optional) – a manual rescaling weight given to each class. If given, has to be a Tensor of size C

size_average (bool, optional) – Deprecated (see reduction).

ignore_index (int, optional) – Specifies a target value that is ignored and does not contribute to the input gradient. When size_average is True, the loss is averaged over non-ignored targets. Note that ignore_index is only applicable when the target contains class indices. Default: -100

reduce (bool, optional) – Deprecated (see reduction).

reduction (str, optional) – Specifies the reduction to apply to the output: 'none' | 'mean' | 'sum'. 'none': no reduction will be applied, 'mean': the sum of the output will be divided by the number of elements in the output, 'sum': the output will be summed. Note: size_average and reduce are in the process of being deprecated, and in the meantime, specifying either of those two args will override reduction. Default: 'mean'

label_smoothing (float, optional) – A float in [0.0, 1.0]. Specifies the amount of smoothing when computing the loss, where 0.0 means no smoothing. The targets become a mixture of the original ground truth and a uniform distribution as described in Rethinking the Inception Architecture for Computer Vision. Default: 0.00.00.0.

Input: Shape (C)(C)(C), (N,C)(N, C)(N,C) or (N,C,d1,d2,...,dK)(N, C, d_1, d_2, ..., d_K) (N,C,d1,d2,...,dK) with $K \geq 1$ ≥ 1 in the case of K-dimensional loss.

Target: If containing class indices, shape ()(), (N)(N)(N) or (N,d1,d2,...,dK)(N, d_1, d_2, ..., d_K)(N,d1,d2,...,dK) with $K \geq 1$ ≥ 1 in the case of K-dimensional loss where each value should be between [0,C][0, C][0,C). If containing class probabilities, same

shape as the input and each value should be between $[0,1]$ $[0, 1]$ $[0,1]$.